

Exercise 3: Visual Encoding

Submission: Sunday, 19th April, 23:59h

Task 1: Variable Encoding

In this task, we will explore different visual encoding strategies and you have to create a scatter plot that can be controlled by at least 4 visual variables (e.g., glyph/shape, size, color hue, brightness, saturation, orientation, pattern/texture) in addition to the two positional ones on the two axes.

- a. Download the file **cleaned_dataset.csv** on OLAT. The dataset contains the following variables:
 - **Area**: area name
 - **Year**: year
 - **Item**: crop type
 - **Yield_hg_ha**: crop yield (hectograms per hectare)
 - **Rainfall_mm**: average annual rainfall (millimeters)
 - **Temp**: average annual temperature (°C)
 - **Pesticides_ton_ha**: total pesticide usage (tons per hectare)
 - **Continent**: Continent of the area
- b. For all attributes, except *Year* and *Area*, create an attribute selection pool (e.g., a set of radio buttons) that allows the user to assign one specific visual encoding method to each attribute separately. In total, you will need 6 different radio button sets to let the user set the visual encoding. (Note: add an additional button for "None", so that you can change the encoding assignment strategy.) 1 point
- c. According to the attribute-encoding pair selected in the selection pool, dynamically draw a scatter plot. Ensure that each visual encoding is assigned to **only one attribute at a time**. Clearly indicate in the legend which attribute is mapped to which visual encoding. 2 points
- d. Try out different encoding assignment strategies, and decide for yourself, and following the lecture notes, on one best solution. Submit the final encoding configuration you consider best together with an explanation of your choice in a .txt or .pdf file. 1 point

Task 2: Interaction

1 point

In this task, you will extend the scatter plot from Task 1 by adding the following interactive filtering components:

- **Year:** Add a slider for the user to select the year. The scatter plot should display only data points from the year selected on that slider widget.
- **Item and Continent:** Add a multi-select widget (e.g., dropdown menus), so that the user can filter the dataset by crop (**Item**) or continent.
- **Yield_hg_ha, Rainfall_mm, Temp, Pesticides_ton_ha:** Add an interval range slider for each of them to filter the dataset between selected minimum and maximum values.

Hand-in

Please hand in your solution in OLAT as a **.zip** file until **Sunday, 19th April, 23:59h**. The **.zip** file should include the following:

- Your documented **python code** with the executable final application in a single file (.py) with a good and useful amount of comments explaining your implementation.

Note: You are **not** allowed to use **other libraries** except for the ones stated in environment.yml.

- Include a few **screenshots**, or a **video**, demonstrating your running application.

Note: Submissions without screenshots will result in a **0.5 point deduction**.

- Required .txt or .pdf file. In case there are additional remarks, include them as a .txt or .pdf file too.
- Please hand-in your own unique solutions. We do not accept copied solutions or solutions created by or with ChatGPT!